

Step by Step Guide: Complete Wireshark Course

A practical follow-along guide for network protocol analysis

How to Use This Guide

This guide follows the sequence of the video lessons from installation through to full attack analysis. Each part lists the tasks to perform in Wireshark, the exact packets or filters referred to in the video, and the observations expected at the end of the task. Timestamps are given so the relevant section of the video can be located quickly.

Work through the parts in order. Pause the video at the start of each part, perform the steps in Wireshark, then resume.

Requirements

- A computer running Windows 64-bit with administrator rights.
- An active internet connection through a wired or wireless adapter.
- A web browser for generating traffic during live capture exercises.
- The sample capture file file1.pcapng, plus the additional capture files used in the later attack analysis sections.
- No previous experience with network analysis is required.

Note on Legal and Ethical Use

Capture traffic only on networks that you own or on which you have written permission to perform analysis. Capturing traffic on a network without authorisation is unlawful in most jurisdictions.

Part 1: Installing Wireshark

1. Open a web browser and go to the Wireshark website at [wireshark.org](https://www.wireshark.org).
2. Select the Windows 64-bit installer from the download page. Other platforms are listed, but this guide uses the Windows installer.
3. Run the downloaded installer once the file has finished downloading.
4. Read the licence agreement and select I Agree to continue.
5. Review the component list. TShark is the command line version of Wireshark and is included by default. Leave all defaults selected.
6. Keep the user guide component selected. It contains reference material and sample packet captures used later.
7. Accept all remaining default settings without changes.
8. At the packet capture driver screen, check the version reported. If the installer states that the same version of the capture driver is already installed, no reinstallation is needed. Select Next.
9. If no capture driver has ever been installed on the machine, allow the installer to install it. USB capture support is not required for this course, so accept the defaults.
10. Select Install and wait for the installation to complete.
11. Select Next, then choose the option to run Wireshark.

Observation: version of Wireshark installed and the capture driver status reported by the installer
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Part 2: The Wireshark Interface

The opening screen shows a welcome message and a list of detected network adapters. Each adapter displays a small graph indicating whether packets are currently moving through it.

Starting a Capture

1. Identify the adapter that is connected to the network. Its graph will show activity.
2. Double click the adapter to begin capturing immediately, or use Capture then Options from the menu.
3. In the capture options window, select the interface and choose Start.

The Three Panes

The capture window is divided into three sections. Understanding the role of each is essential before any analysis begins.

- Packet list, at the top. Displays one row for each packet, showing the packet number, time, source, destination and protocol.
- Packet details, in the middle. Displays the protocol layers of the selected packet, such as the frame, Ethernet, and any encapsulated protocol. Each layer can be expanded for more detail.
- Packet bytes, at the bottom. Displays the raw contents of the packet in hexadecimal, which is far easier to read than binary. As an example, six pairs of FF in the destination field indicate a broadcast address.

Reference Resources Mentioned

- Wireshark User Guide, which contains detailed usage information and sample packet captures.
- Wireshark Wiki, an extensive source of protocol and capture information.
- SharkFest, the annual Wireshark conference, which publishes presentations and workshop material.

Part 3: Reviewing Your First Packet Capture

1. Open Wireshark and select File then Open.
2. Locate the file named file1.pcapng, select it and choose Open.
3. Look at the packet list at the top. Note the large number of ARP packets. ARP stands for Address Resolution Protocol.
4. Examine one ARP row. It shows a source, such as a Netgear MAC address, sending to a broadcast destination.
5. Scroll through the packet list and identify other protocol types. Locate the HTTP traffic. HTTP is carried in clear text, so its contents are readable.
6. Select an HTTP packet and look at the packet details pane. The layers appear in order as frame, Ethernet, IP, then TCP.
7. Look at the packet bytes pane at the bottom. These hexadecimal values are the actual data on the wire.

Observation: protocols identified in file1.pcapng and the layer order shown for an HTTP packet

Part 4: Network Communication, DNS and ARP

Concepts to Note Before the Practical Steps

- Clients normally use source ports above 1023. These are described as transient or ephemeral because a client does not stay connected for long.
- Servers normally use ports below 1023. These are described as reserved or privileged ports and remain open to accept connections, which is why they are frequent targets of attack.
- DNS translates a human readable domain name into an IP address that a machine can process.
- An IPv4 address is written as four decimal numbers known as dotted octets, each representing eight bits. Humans read 192.168.0.1, while the machine processes a string of ones and zeros. Hexadecimal is used as a readable middle ground.
- ARP maps an IP address to a physical MAC address. The IP address delivers the data to the correct network, and the MAC address delivers it into the correct machine.

Practical Steps in file1.pcapng

1. With file1.pcapng open, scroll the packet list and locate DNS traffic.
2. Go to packet 1093, which contains a DNS query for www.udemy.com.
3. Check the ports on this packet. The source port is above 1023 and the destination port is 53. This is the standard pattern of client to server communication.
4. Expand the Domain Name System query section in the packet details pane to view the query fields.
5. Move to the following packet, which is the DNS response. Record the IP address returned, which is 151.101.24.175.
6. Now locate the ARP exchange in the packet capture. Find the ARP request asking for the MAC address that belongs to 192.168.0.7.
7. Select the matching ARP reply, at packet 3714, which supplies the MAC address for that host.
8. Select the ARP packet and expand the ARP section to view the sender and target MAC and IP addresses.

Observation: DNS query name, resolved IP address, source and destination ports, and the MAC address returned by ARP

Part 5: Core Protocols, ICMP, UDP and TCP

Concepts to Note Before the Practical Steps

- ICMP, the Internet Control Message Protocol, carries control messages within a network. Protocols with no acknowledgement mechanism of their own rely on it. The most familiar use is ping, which sends an echo request and expects an echo reply.
- ICMP has many types. Type 3 means destination unreachable, and its subtypes, known as codes, explain the reason. Code 3 means the port is unreachable, and code 13 means the packet was administratively prohibited.
- UDP, the User Datagram Protocol, is connectionless and unreliable. It is built for speed and gives no guarantee of delivery, in the same way that a posted letter carries no confirmation of arrival. It suits applications such as streaming.
- TCP, the Transmission Control Protocol, carries most network traffic. It provides reliability and guaranteed delivery through acknowledgements, and it is full duplex, meaning both sides can transmit at the same time. The cost of this reliability is additional overhead and lower speed than UDP.

Practical Steps in file1.pcapng

1. Select the capture window and press Ctrl and End together to jump to the bottom of the capture.
2. Locate packet 8217, an ICMP echo request, and packet 8218, the matching echo reply. This is a ping from host 192.168.0.187 to the web server.
3. Expand the ICMP section in the packet details pane. Confirm that the echo request is type 8 and the echo reply is type 0.
4. Use Go, then Go to Packet, and enter 1241 to move directly to that packet.
5. Examine packet 1240 first. It is a DNS query sent over UDP to port 53.
6. Examine packet 1241. It is an ICMP type 3 response with code 3, indicating that the destination port is unreachable.
7. Expand the UDP section on packet 1240 and note how few fields it contains. UDP is connectionless and carries no delivery guarantee, so its header is minimal.
8. Go to packet 1239, which uses TCP. Expand the TCP section and scroll to the flags. Confirm that the ACK flag is set, which is part of the reliability mechanism of TCP.

Observation: ICMP types recorded, the ICMP type 3 code and its meaning, and the TCP flag observed

Part 6: TCP/IP Encapsulation and Headers

Concepts to Note Before the Practical Steps

- The internet was built on openness and trust, at a time when it was far smaller than today. Many of those trusting protocols are still in use, which is one reason so many attacks succeed against them.
- At each layer of communication, information is added to the data in the form of a header that identifies the data to the receiver. Reading these headers is how an analyst determines what is happening inside a packet.
- The process of wrapping data inside each successive layer is called encapsulation. The OSI model illustrates this, with each layer adding a header, down to the physical layer of ones and zeros.
- In Wireshark this appears as the frame, then Ethernet, then IP, then the encapsulated protocol such as TCP.

Practical Steps in file1.pcapng

1. Select packet 3, which is an ICMP packet. In the packet details pane confirm the layer order of frame, Ethernet, IP and ICMP.
2. Expand the IP section. Select the IP layer and observe that the whole IP header is highlighted in the bytes pane in hexadecimal.
3. Select the version field. The value 4 marks the start of the IP header.
4. Locate the protocol field. A decimal value of 1 shows that ICMP is the encapsulated protocol.
5. Select packet 5, which is a UDP packet. Confirm that it is also encapsulated inside an IP header, and that the protocol field shows decimal 17, which is hexadecimal 11.
6. Expand the UDP section and note the short, simple header.
7. Select packet 17, which is a TCP packet directed to port 21. Expand the IP section and confirm the protocol field shows decimal 6 for TCP.
8. Expand the TCP section. Note the larger header, which carries sequence numbers, acknowledgement numbers and control bits.
9. Locate the flags within the TCP header and identify the SYN flag, which is the request to open a connection.

Protocol numbers at the ninth byte offset of the IP header:

Protocol	Decimal	Hexadecimal
ICMP	1	01
TCP	6	06
UDP	17	11

Part 7: Following Streams

A stream is the complete conversation between two hosts. Wireshark can reassemble a stream and display it as readable text, which is one of the most useful features of the tool.

Following a UDP Stream

1. With file1.pcapng open, locate the UDP packets that appear early in the capture.
2. Select packet 59, which uses SSDP, the Simple Service Discovery Protocol.
3. Look at the packet details pane. It refers to hypertext transfer protocol because SSDP uses similar parameters. Locate the user agent field.
4. Right click the packet, then select Follow, then UDP Stream.
5. Read the reassembled conversation. It shows that the request came from Google Chrome version 52 running on Windows.
6. Record this finding. An attacker who learns the exact browser version can search for vulnerabilities affecting that version.

Following a TCP Stream

1. Go to packet 17. This is a TCP SYN packet directed to an FTP server.
2. Examine packets 18 and 19, which complete the exchange. Together the three packets form the three way handshake of SYN, SYN ACK, then ACK. No data flows until this handshake is complete.
3. Right click the packet, then select Follow, then TCP Stream.

4. Read the reassembled stream. Text shown in one colour is what the server sent, and text in the other colour is what the client sent.
5. Identify the username and password in the stream. FTP is built on the principle of trust, so credentials are carried in clear text.
6. Identify the server version and the name of the file transferred, which is mail.gif.
7. Note that server version information of this kind is valuable to an attacker and should be suppressed on a production system wherever possible.

Observation: browser and version identified from the UDP stream, and the credentials, server version and filename identified from the TCP stream

Part 8: Performing Your First Live Capture

Understanding the Capture Options

- Recent versions of Wireshark allow more than one interface to be selected for a single capture.
- All detected interfaces are listed when the tool opens. If an expected interface is missing, the cause is usually a driver problem or host based protection software on the machine.
- Promiscuous mode turns off MAC address filtering on the adapter.
- The Output tab allows the capture to be saved in different formats, including pcap, which can be read by other tools such as tcpdump and intrusion detection systems such as Snort.
- A file size limit can be set. Wireshark then creates a new file automatically when the limit is reached, which supports continuous monitoring.
- The Name Resolution tab controls whether names are resolved in place of raw addresses.

Capturing an Encrypted Session

1. Start Wireshark and confirm that the interface connected to the internet is listed. In the video this is the wireless interface.
2. Select Capture then Options. Confirm that the interface shows activity on its graph.
3. Select the interface and open the options. Open the Name Resolution tab and clear the check box for resolving MAC addresses. Without this, vendor names appear instead of MAC addresses.
4. Select Start to begin capturing. Confirm that MAC addresses now appear in place of vendor names.
5. Open a web browser and visit a website.
6. Return to Wireshark and stop the capture.
7. Search the packet list for the DNS query for that site. In the video this is packet 106.
8. Locate the query and its response, followed by the start of the TCP connection.
9. Right click the SYN packet and select Follow, then TCP Stream.
10. Note that the data is encrypted and cannot be read, but that some information is still visible, such as the SSL certificate details.

Capturing an Unencrypted Session

1. Start a new capture on the same interface.
2. Visit a site that is known to use HTTP rather than HTTPS.

3. Stop the capture and locate the DNS query and response again.
4. Locate the SYN packet for the TCP connection to port 80, which confirms HTTP.
5. Right click the SYN packet and select Follow, then TCP Stream.
6. Read the full exchange. Identify the client request, the server response and any cookies.
7. Check whether the server is disclosing its own version information. A server that does not leak this is the better security position, since disclosed versions expose known vulnerabilities.
8. Use the display option within the stream window to show only the server side or only the client side of the exchange.

Observation: information still visible in the encrypted session, compared with the information visible in the unencrypted session

Part 9: Promiscuous Mode and an Analysis Method

Checking Promiscuous Mode

1. Select Capture then Options.
2. Locate the setting to enable promiscuous mode on all interfaces and confirm its state.
3. Note the effect. On a switched network you will see only traffic to and from your own machine, together with broadcast traffic, whatever this setting is. Promiscuous mode is useful when a hub or a device is forwarding all network traffic to the machine, in which case all traffic becomes visible.
4. Note the second use case. Older versions of Wireshark required a wireless adapter not to be in promiscuous mode in order to see Ethernet traffic. Current versions do not have this limitation. If an older version is capturing nothing, this setting is a likely cause.

Applying a Repeatable Method

1. Start a new capture on the interface connected to the internet.
2. Generate traffic by visiting a website in the browser, then stop the capture.
3. Go to the top of the capture at packet 1. Expect some UDP and ARP traffic, which is normal on any network.
4. Work down the list looking for UDP traffic, specifically the DNS query for the site you visited.
5. Once the DNS query is found, read the response immediately after it to obtain the IP address of the site.
6. Follow the connection to that IP address. Identify the three way handshake on port 80, which confirms HTTP.
7. Right click the packet following the query and select Follow, then TCP Stream.
8. Read the server information in the stream. In the video the server identifies itself as Microsoft IIS 8.0.
9. Clear the filter and continue examining the capture for anything unusual.
10. Note that Wireshark may flag spurious retransmissions. These are frequently just background noise on the network, much of it produced by data mining, which is why advertisements relate to recent searches.
11. Repeat the same method on a different object in the capture, such as a JPEG file, and follow its TCP stream to see what further information is exposed.

Summary of the method: locate the DNS query, read the response for the IP address, follow the connection, follow the stream, then record what the server has disclosed.

Part 10: Saving, Exporting and Merging Capture Files

File Formats

- pcapng is the current default format and holds the most detail.
- pcap is the most versatile format. It allows the capture to be opened by command line tools such as tcpdump and imported into tools such as Snort. Use this format when the file must be shared across tools.

Saving a Capture

1. Confirm the interface is connected to the internet, then select Capture then Options and start the capture.
2. Generate traffic by opening the browser and visiting a website.
3. Return to Wireshark and stop the capture.
4. Select File then Save As. Choose pcap from the format list to keep the file usable in other tools.
5. Give the file a descriptive name that identifies the site or the activity captured.

Isolating and Saving Part of a Capture

1. Scroll the saved capture and locate the DNS query for the site you visited.
2. Right click the packet and select Follow, then TCP Stream.
3. Read the server response and record the server version disclosed.
4. Save only this component as a separate file, named to describe its contents, for example a name that identifies the web server traffic for that site.
5. Note the purpose. Isolated captures are used for intrusion analysis, for training material and as evidence, because they contain only the relevant packets.

Merging Capture Files

1. Start a capture, generate some browser traffic, then stop the capture.
2. Select File then Save and give the file a name such as website traffic.
3. Select File then Merge.
4. Locate the saved file and open it, then merge it with file1.pcapng.
5. Scroll through the result and confirm that packets from both sources are present.
6. Save the merged file with a name that records what was combined.
7. Remove unwanted content, such as the ARP traffic carried in from file1.pcapng, by locating the section you want to keep, following its stream and saving only that section under its own name.
8. Note the value of this. Merged and trimmed captures build a library of training material and produce clean evidence files with the noise removed.

Importing Hexadecimal Data

If another tool produces output in hexadecimal and saves it as a text file, Wireshark can import and read that file through its graphical interface, which is far easier to interpret than raw binary data.

Part 11: Filtering

Filtering replaces the slow method of scrolling manually through a capture. Wireshark provides two distinct types of filter.

- Capture filters control which packets are captured in the first place. They can be built to watch for specific types of attack, so that any traffic matching them becomes evidence of a possible intrusion. Packets that do not match are never saved.
- Display filters control which packets are shown on screen. Every packet remains in the file. These are used to narrow the view to a protocol, an address or a single conversation.

Applying Display Filters

1. Open your capture file in Wireshark.
2. Type `arp` into the filter bar and select the arrow to apply it. Only ARP packets remain visible.
3. Clear the filter by selecting the X at the end of the filter bar.
4. Apply the filter `icmp` and confirm that only ICMP traffic is shown.
5. Apply the filter `tcp` and confirm that only TCP traffic is shown.
6. With the TCP filter applied, examine packet 17, which shows the start of the three way handshake to an FTP server, and packet 20, which identifies the FTP server version.
7. Note that any protocol name can be entered in the filter bar in the same way.

Applying a Capture Filter

1. Select Capture then Options.
2. Locate the capture filter field.
3. Enter the filter `udp port 53` to capture DNS traffic only.
4. Start the capture and generate traffic in the browser.
5. Return to Wireshark and confirm that only DNS packets have been captured.
6. Note the limitation. A capture filter restricts what is saved, so any other packet of interest is lost permanently. Use capture filters only where you are certain the other protocols are not needed.

Filters used in this course:

Filter	Type	Purpose
<code>arp</code>	Display	Show ARP packets only
<code>icmp</code>	Display	Show ICMP packets only
<code>tcp</code>	Display	Show TCP packets only
<code>udp port 53</code>	Capture	Capture DNS traffic only
<code>tcp.flags.push == 1</code>	Display	Show only packets that carry data
<code>tcp.flags.syn == 1 and tcp.flags.ack == 1</code>	Display	Show replies that indicate an open port

Part 12: Normal and Abnormal Traffic

TCP Flags

TCP reliability is controlled by six flags, also called control bits. Recognising them is what allows normal traffic to be separated from abnormal traffic.

Flag	Name	Meaning
URG	Urgent	The packet carries urgent data
ACK	Acknowledgement	A packet has been received
PSH	Push	The packet carries data
RST	Reset	Tear down the connection
SYN	Synchronise	Request to open a connection
FIN	Finish	Request to close the connection

Connection Setup and Close

- The three way handshake opens a connection in three steps. SYN requests the connection, SYN ACK acknowledges the request and opens the receiving side, and the final ACK acknowledges that synchronisation. Both sides must open their own side, which is what makes TCP full duplex.
- Closing takes four steps, because each side closes independently and each must be allowed to finish sending. The sequence is FIN, ACK, then FIN, ACK.
- In the sample capture the handshake appears in packets 18 to 22, and the four step close appears in packets 23 to 26.
- A reset is used when something goes wrong and the connection must be torn down. A large number of resets in a capture indicates an abnormal condition, commonly an attempt to connect to a service that is not running.

Recognising a Crafted Packet

A packet with the FIN, PSH and URG flags set at the same time is abnormal, because it carries data in the same packet that closes the connection. This combination is the signature of the Xmas scan performed by the Nmap scanning tool.

Practical Steps in the Attack Capture File

1. Open the capture file used in this section of the video.
2. Look for the warning that a duplicate IP address has been detected. This means two machines on the network are claiming the same address, which creates confusion about where data should be delivered.
3. Record this as an indicator of ARP spoofing, also called ARP poisoning, in which an attacker pretends to be both machines. This is the classic man in the middle attack.
4. Scroll after the spoofing activity and note the large number of reset packets. SYN packets followed immediately by resets are the signature of a port scan, in which an attacker tests many ports to find which ones respond.
5. Apply the display filter `tcp.flags.push == 1` to show only packets that carry data.
6. Examine the remaining packets. Those with FIN, PSH and URG set together confirm the Xmas scan.
7. Go to packet 10295 and read the info column. It shows that a version of BIND, a DNS server, was returned.
8. Follow the TCP stream for that packet. The server discloses its DNS server version as `dnsmasq 2.39`.
9. Clear the filter and apply a filter for HTTP traffic.
10. Examine the HTTP responses. The server discloses its Apache version and its OpenSSL version. This is excessive disclosure and provides an attacker with direct attack vectors.

Observation: indicators of ARP spoofing and port scanning found, and the software versions disclosed by the target

Part 13: Analysing a Compromised Host

1. Open the second attack capture file and apply the same method used in the previous part.
2. Examine packet 4. The protocol is NBSS, the NetBIOS Session Service, a local area network protocol used by Windows for name communication.
3. Record the two host names visible in this packet, which are Vulcan21 and external DMC. These hosts are communicating on a Windows network.
4. Apply the display filter `tcp.flags.syn == 1` and `tcp.flags.ack == 1` to show every packet with both the SYN and ACK flags set. These replies identify open ports.
5. Record the open ports found, such as port 80.
6. Apply the display filter `tcp.flags.push == 1` to reduce the capture to packets that carry data, which makes suspicious activity easier to find.
7. Examine packet 59. It shows FTP data on port 80, which is not the expected port for FTP and is therefore suspicious.
8. Follow the TCP stream for packet 59. It shows an anonymous user logging in and transferring a file.
9. Continue through the capture looking for signs of compromise, particularly attempts to obtain a shell on the server. Look for references to `bin sh`, which indicate an attempt to execute shell commands.
10. Look for a returned value of `uid` equals 0. This confirms root level access and therefore a compromised machine.
11. Examine packet 1211. The attack itself failed, but the response disclosed a very old version of Microsoft IIS 5.0, which is a serious finding for the client.
12. Identify the directory traversal attempt, indicated by the encoded characters `%c0%af` in the request.
13. Examine packet 1495. It confirms a successful compromise. The attacker obtained a shell, verified that the host was running Microsoft Windows 2000, then exited.

Conclusion: summarise the attacks identified in this capture, the evidence for each and the state of the target host at the end of the capture

Appendix A: Packet Reference from the Video

The packet numbers below refer to the sample capture files used in the video lessons.

Packet	Content	Part
3	ICMP packet used to examine the IP header	Part 6

Packet	Content	Part
5	UDP packet showing protocol value 17	Part 6
17 to 19	TCP three way handshake to an FTP server	Parts 6 and 7
20	FTP server version disclosed	Part 11
59	SSDP over UDP, browser version disclosed	Part 7
1093	DNS query for www.udemy.com	Part 4
1239	TCP packet with the ACK flag set	Part 5
1240 and 1241	DNS query over UDP and ICMP type 3 code 3 reply	Part 5
3714	ARP reply supplying a MAC address	Part 4
8217 and 8218	ICMP echo request and echo reply	Part 5
10295	BIND version disclosed, dnsmasq 2.39	Part 12
4	NBSS host names in the attack capture	Part 13
1211	Failed attack disclosing Microsoft IIS 5.0	Part 13
1495	Confirmed compromise, shell on Windows 2000	Part 13

Appendix B: Checklist

Use this checklist to confirm that each part of the course has been completed.

- Wireshark installed and opened successfully
- Three panes of the interface identified
- file1.pcapng opened and protocols identified
- DNS query and response located, resolved IP recorded
- ARP request and reply located, MAC address recorded
- ICMP echo request and reply identified by type
- ICMP type 3 destination unreachable located
- UDP and TCP headers compared
- Encapsulation confirmed using the IP protocol field
- UDP stream followed and browser version recorded
- TCP stream followed and FTP credentials recorded
- Live capture performed on an encrypted site
- Live capture performed on an unencrypted site
- Capture saved in pcap format
- Section of a capture isolated and saved separately
- Two capture files merged
- Display filters applied for arp, icmp and tcp
- Capture filter applied for udp port 53
- Six TCP flags identified and their meanings recorded
- ARP spoofing indicator located in the attack capture
- Port scan and Xmas scan signatures identified
- Disclosed server versions recorded
- Compromised host evidence identified and summarised